

Section -5 Terms of Reference (ToR)

1. Introduction

The Directorate General of Shipping (DGS), an attached office of the Ministry of Ports, Shipping and Waterways, Govt. of India, deals in matters relating to merchant shipping. The DGS deals with all matters concerning the Maritime Administration, Maritime Education and Training, development of Shipping Industry and other related subjects.

The Directorate General of Shipping (DGS), Ministry of Ports, Shipping and Waterways, intends to engage a competent Consultancy Firm to provide professional services for the Design, Development, and Implementation of an Internal Digital Platform for Managing India's IMO-Related Coordination and Strategic Engagement.

The consultancy support is required for system design, process mapping, digital workflow formulation, requirements finalization, and preparation of detailed technical and functional documentation, along with advisory support for implementation, change management, and capacity building.

This assignment will contribute significantly to strengthening India's representation, preparedness, and participation in global maritime forums and aligns with Government of India's priorities on maritime safety, environmental protection, regulatory modernization, and digital governance.

1.1 Background – International Cooperation, DGS

The International Cooperation (IC) Branch of the Directorate General of Shipping is the nodal unit responsible for coordinating India's engagement with international maritime bodies, including the International Maritime Organization (IMO), International Labour Organization (ILO), International Maritime Satellite Organization (IMSO), and various bilateral and multilateral working groups.

Given the increasing volume of agenda papers, technical submissions, deliberations, inter-ministerial inputs, and committee-level decisions, the IC Division requires a centralized digital system to streamline internal collaboration, document handling, paper preparation, feedback consolidation, and delegation management.

To support this growing requirement, the IC Division seeks to onboard a consultancy and technology partner for establishing a IMO Strategic Engagement Platform, facilitating structured workflows and enhancing India's institutional readiness.

1.2 Overview of the International Maritime Organization (IMO)

The International Maritime Organization (IMO) is a specialized agency of the United Nations responsible for regulating shipping, enhancing maritime safety, and preventing marine pollution. The IMO consists of:

- I The Assembly – the highest governing body, meeting every two years.
- II The Council – comprising 40 elected Member States, functioning as the governing body between Assembly sessions
- III Five main Committees, which undertake the technical work of the Organization:
 - a) Maritime Safety Committee (MSC)
 - b) Marine Environment Protection Committee (MEPC)
 - c) Legal Committee
 - d) Technical Cooperation Committee

Engagement Platform

e) Facilitation Committee

IV The MSC and MEPC are further supported by seven Sub-Committees open to all Member States:

- a) Human Element, Training and Watchkeeping (HTW)
- b) Implementation of IMO Instruments (III)
- c) Navigation, Communications and Search & Rescue (NCSR)
- d) Pollution Prevention and Response (PPR)
- e) Ship Design and Construction (SDC)
- f) Ship Systems and Equipment (SSE)
- g) Carriage of Cargoes and Containers (CCC)

V The Committees & Sub-Committees are further supported by:

- a) Intersessional Working Groups
- b) Joint Working Groups
- c) Correspondence Groups
- d) Expert Groups

India, as a founding Member State, actively participates in IMO committees, sub-committees, working groups, and correspondence groups. India contributes significantly to global maritime policy, safety regulations, environmental initiatives, and new technology-driven reforms.

To further strengthen India's participation, improve inter-agency coordination, and ensure timely, high-quality submissions, there is a need for a dedicated internal digital platform for managing all IMO-related activities, documentation, workflows, and preparatory processes.

2. Objectives of the Assignment

The primary objective of this assignment is to support the Directorate General of Shipping (DGS), particularly the International Cooperation (IC) Branch, in conceptualizing, designing, and facilitating the development of a unified IMO Strategic Digital Platform that strengthens India's institutional capacity for effective engagement with the International Maritime Organization and other global maritime bodies.

The key objectives of the assignment include:

2.1 Strengthen India's Preparedness and Participation in IMO/ILO

- Enable structured internal coordination on MSC, MEPC, and other IMO/ILO committees, sub-committees, Intersessional Working Groups, Joint Working Groups, Correspondence Groups, and Expert Groups
- Enable structured internal coordination on other joint working groups, expert groups, etc as may be set up by DGS/MoPSW related to International Cooperation Matters
- Improve quality, accuracy, and timeliness of India's positions, interventions, and submissions
- Ensure a standard workflow for processing agenda items, technical papers, and policy inputs

2.2 Establish a Centralized Digital Collaboration Framework

- Design a cloud-native platform for seamless collaboration across DGS, MMDs, MoPSW, and other stakeholders
- Facilitate document management, agenda wise deliberations, feedback collation, and meeting preparation

2.3 Streamline End-to-End IMO Documentation & Workflow Processes

- Digitize the entire workflow—from agenda distribution to drafting, review, approval, and submission.
- Standardize processes for Correspondence Groups, Working Groups, and inter-ministerial consultations.

2.4 Enhance Transparency, Traceability, and Knowledge Retention

- Maintain a complete audit trail of revisions, discussions, comments, and approvals.
- Centralize archival of IMO documents, country positions, meeting minutes, and intervention statements.
- Enable long-term knowledge continuity within DGS and support institutional memory.

2.5 Support Digital Transformation and Compliance

- Align the system with Government of India's digital governance standards, MeitY cloud guidelines, and data security norms.
- Improve efficiency, reduce duplication, and enhance the accuracy of submissions through digital automation.

2.6 Build Capacity and Support Change Management

- Strengthen capabilities of DGS and stakeholder officials through training, workshops, and user familiarization.
- Recommend process improvements and governance structures for sustained long-term use of the platform.

3. Scope of Services

The selected System Integrator (SI) shall be responsible for the **end-to-end design, development, implementation, cloud hosting, testing, deployment, training, and operations & maintenance** of the proposed IMO Strategic Engagement Platform for the Directorate General of Shipping (DGS).

The scope includes all activities required to deliver a secure, scalable, cloud-native platform along with sustained post-implementation support.

3.1 Requirement Gathering & Finalization

- i. Conduct detailed requirement-gathering workshops with DGS, International Cooperation Division, MMDs, MoPSW, and other identified stakeholders.
- ii. Study existing processes, legacy systems, documentation practices, and workflows.
- iii. Clearly document functional workflows, user roles, approval hierarchies, and integrations.
- iv. Capture functional, technical, security, performance, and compliance requirements.
- v. Finalize requirements in consultation with DGS and obtain formal sign-off.
- vi. Maintain a Requirement Traceability Matrix (RTM) throughout the project lifecycle.

3.2 System Architecture & Design

- i. Design a cloud-native, modular, and scalable architecture aligned with MeitY guidelines.
- ii. Prepare and submit for approval:
 - Functional Requirements Specification (FRS)
 - Software/System Requirements Specification (SRS)
 - High-Level Design (HLD)
 - Low-Level Design (LLD)
 - Data Models and ER Diagrams

- UI/UX wireframes and user journeys
- iii. Define role-based access control, workflows, audit trails, dashboards, and reporting structures.

The System Integrator (SI) shall design, implement, and finalise a secure, scalable, cloud-native, and modular system architecture. The architecture shall be approved by the Directorate General of Shipping (DGS) prior to development and shall meet the following **minimum requirements**.

- A. The frontend shall provide a responsive, secure, and user-friendly interface for all user roles.
 - i. Web-based application accessible through standard modern browsers (Chrome, Edge, Firefox).
 - ii. Responsive design supporting desktop, tablet, and mobile access.
 - iii. Role-based UI rendering (Admin, Delegation Leader, Coordinator, Member, Viewer).
 - iv. Support for accessibility guidelines (WCAG 2.1 Level AA).
 - v. Secure session management and inactivity time-outs.
 - vi. Internationalization support (future-ready).
 - vii. Separation of UI and business logic (thin client architecture).
 - viii. Protection against common vulnerabilities (XSS, CSRF, clickjacking).
- B. The backend shall handle business logic, workflows, data processing, and security enforcement.
 - i. Service-oriented or microservices-based architecture.
 - ii. Stateless application design for scalability.
 - iii. Role-based access enforcement at API level.
 - iv. Workflow engine to support multi-level approvals and task routing.
 - v. Centralized exception handling and logging.
 - vi. High availability and horizontal scalability.
- C. The middleware shall enable secure and controlled integration between system components and external systems.
 - i. API-first architecture using RESTful APIs.
 - ii. Secure API Gateway with authentication, throttling, and logging.
 - iii. Support for future integration with IMO, ILO, IMSO, or other Government systems.
 - iv. Message-based integration (where required) for asynchronous processing.
 - v. Versioned APIs to ensure backward compatibility.
- D. The data layer shall ensure integrity, availability, confidentiality, and long-term retention of critical information.
 - i. Enterprise-grade relational database.
 - ii. Data normalization and indexing for performance.
 - iii. Support for structured and unstructured data (documents, feedback, logs).
 - iv. Role-based data access control.
 - v. Encryption at rest and in transit.
 - vi. Point-in-time recovery and automated backups.
 - vii. Archival and retention policies aligned with Government norms.
- E. Logging, Monitoring & Audit Architecture
 - i. Centralized logging across all components.
 - ii. Real-time monitoring of application and infrastructure health.
 - iii. Audit trails for document changes, approvals, and user actions.
 - iv. Dashboard for system administrators.
 - v. Log retention as per Government norms.

F. Architecture Governance

- i. The SI shall submit detailed architecture diagrams and documentation.
- ii. Any deviation from the approved architecture shall require prior written approval from DGS.
- iii. Architecture shall be reviewed at design, pre-Go-Live, and post-Go-Live stages.

3.3 Design & Development of the Application

The System Integrator (SI) shall be responsible for the **end-to-end design, development, configuration, customization, and implementation** of the complete **IMO Strategic Platform**, in accordance with the approved Functional Requirements Specification (FRS), Software/System Requirements Specification (SRS), and system design documents.

The SI shall ensure that the application is developed using a modular, scalable, and cloud-native architecture, enabling ease of maintenance, future enhancements, and seamless integration with other Government systems, international portals, and digital platforms as required by DGS.

The SI shall implement all **business workflows, approval hierarchies, collaboration mechanisms, and reporting frameworks**, ensuring accuracy, traceability, and compliance with Government usability, accessibility, and security standards.

The platform shall include, but not be limited to, the following integrated functional modules and capabilities:

A. Committee, Sub-Committee, Working Group & Meeting Management

- i. Configuration and management of IMO Assembly, Council, Committees, Sub-Committees, Working Groups, and other international forums (including ILO, IMSO, and bilateral/multilateral engagements).
- ii. Creation, scheduling, and management of meetings with participant mapping and role assignment.
- iii. Agenda creation and assignment at meeting, committee, and group levels.

B. Agenda, Document & Version Management

- i. Secure upload, categorization, and storage of agenda papers, working documents, submissions, and reference materials.
- ii. Document tagging based on committee, agenda item, source, and meeting.
- iii. Version control with complete audit trail and time stamping of all uploads and revisions.
- iv. Secure in-system document viewing and controlled downloads.

C. Collaboration, Feedback & Deliberation Management

- i. Structured user-wise and group-wise feedback submission for agenda items and documents.
- ii. Commenting, interventions, and deliberation notes with role-based visibility.
- iii. Consolidation of inputs at multiple levels, enabling informed decision-making.
- iv. Defined review workflow:
Member → Group Leader → Delegation Leader → IC Division.

D. Task Allocation & Workflow Automation

- i. Creation and assignment of tasks linked to agenda items, documents, and meetings.